Payton Johncour

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EDUCATION

Orange Coast College - Costa Mesa, CA | GPA: 3.8 | January 2024 - Present
Candidate for Associate of Science Degrees in Computer Science, Mathematics and Physics
Transferring Spring 2027 for Bachelor of Science in Computer Engineering, Expected Graduation: May 2029
Related Coursework: Data Structures and Algorithms, Object-Oriented Programming (C++), Python Programming

TECHNICAL SKILLS

Languages: C++, Python, Linux Shell (Bash)
Software and Frameworks: OpenCV, PyTorch, Ultralytics YOLO, ROS
Hardware and Systems: Raspberry Pi, Arduino, Microcontrollers, Sony IMX500 Sensor, Stepper and Servo Motors, 3D Printing
CAD Design: SolidWorks, Fusion 360

PROJECTS & EXPERIENCE

Autonomous Computer Vision and Robotics Waste Sorting System | Project Lead | Fall 2025 – Spring 2026
Giles T. Brown Research Symposium

- Fabricated a fully autonomous device capable of sorting trash, compost, and recycling with 80% classification accuracy, eliminating the need for manual sorting later in the recycling process
- Implemented an event-based control loop coordinating ultrasonic sensor inputs and motor outputs to manage multi-level system synchronization
- Trained a YOLOv11 object detection model using PyTorch on environment-specific datasets to achieve 80% classification accuracy across three distinct waste streams
- Optimized the model for edge deployment by converting the best version of the model into a compatible format for on-sensor processing with the Sony IMX500 hardware, reducing latency by 66%
- Implemented a communication pipeline linking a Raspberry Pi and Arduino via serial protocol, parallelizing Python and C++ execution for automated motor control

OC Robotics | Embedded Software Developer | Fall 2025 - Present
OC Robotics Club, University Rover Challenge

- Designed C++ motor control software to implement smooth acceleration and deceleration curves, preventing mechanical wear and joint strain during rapid velocity changes
- Integrated ROS (Robot Operating System) nodes to bridge communication between the high-level autonomy stack and the Arduino microcontroller hardware layer
- Managed codebase integration via GitHub, utilizing issue tracking to systematically identify, assign, and resolve software bugs alongside team members

EXTRA CURRICULARS

Micro Mouse @ OCC | Club Founder | Spring 2026 - Present

- Established and organized a new campus organization to prepare student teams for autonomous robotics competitions, growing active membership to 20+ students
- Facilitate weekly technical workshops covering microcontroller programming, sensor integration, and maze-solving algorithms to build member engineering competencies

Hackathon Best Overall Project $500 Scholarship | October 2025